

Full Reference Manual

Laser Scanner Metrology System

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Audience: Test Engineers

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1. Introduction

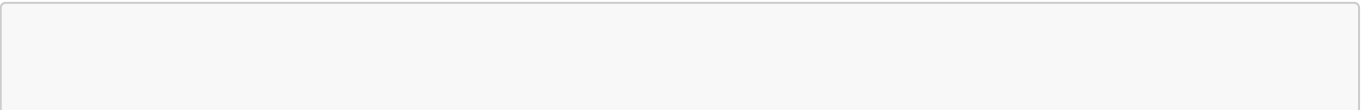
1.1 Purpose

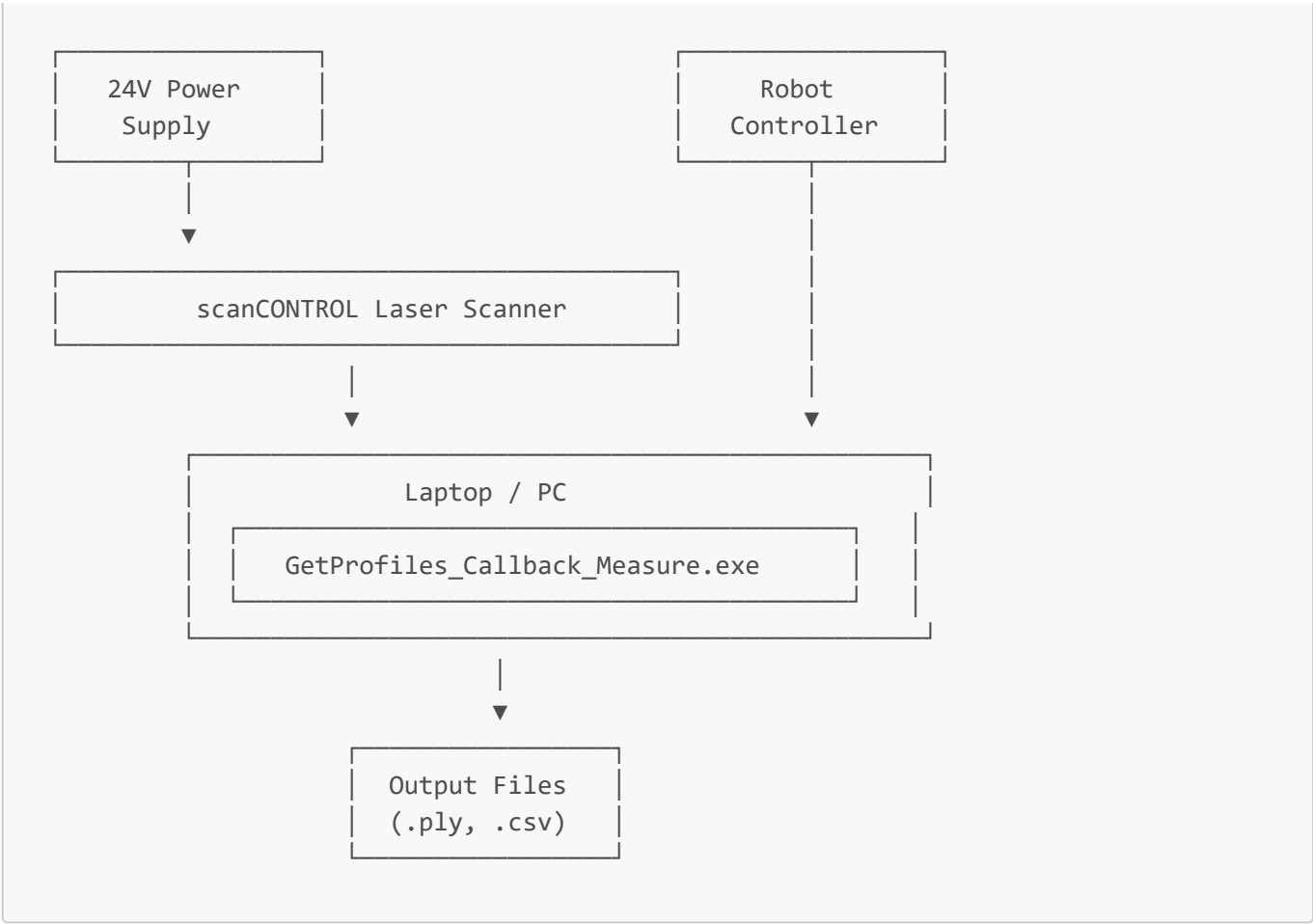
The Laser Scanner Metrology System is an automated 3D surface inspection tool designed to detect and measure surface defects on test specimens. It works in coordination with a robot controller to scan workpieces and generate detailed measurement reports.

1.2 What This System Does

- **Detects** surface defects such as grooves, cuts, scratches, and holes
- **Measures** the area of each detected defect in square millimetres (mm²)
- **Generates** 3D point cloud files for visual inspection
- **Logs** all measurements to CSV files for data analysis

1.3 System Overview





2. System Requirements

2.1 Hardware Requirements

Component	Specification
Laser Scanner	Micro-Epsilon scanCONTROL series
Power Supply	24V DC for scanner
Computer	Windows 10/11 PC with Ethernet port
Network	Direct Ethernet connection to scanner
Robot	Industrial robot with TCP/IP communication capability

2.2 Software Requirements

Software	Purpose	Where to Get It
scanCONTROL Configuration Tools	Laser alignment and setup	Micro-Epsilon website
GetProfiles_Callback_Measure.exe	Main measurement application	Provided with this system
GetProfiles_Callback_Measure.pdb	Debug symbols (keep with .exe)	Provided with this system

Software	Purpose	Where to Get It
LLT.dll	Scanner driver library	Micro-Epsilon SDK
CloudCompare	3D visualisation	https://www.cloudcompare.org/

2.3 File Organisation

Your folder structure should look like this:

```
SDK42/
├── LLT.dll                                ← Scanner driver (MUST be here)
├── examples/
│   ├── GetProfiles_Callback_Measure/
│   │   ├── GetProfiles_Callback_Measure.exe    ← Main application
│   │   ├── GetProfiles_Callback_Measure.pdb    ← Debug symbols
│   │   └── Session_YYYY-MM-DD_HH-MM-SS/        ← Output folders (created
│   │                                           automatically)
└──
```

 **Critical:** The `LLT.dll` file must be in the **parent folder** of the executable, not in the same folder.

3. Hardware Setup

3.1 Scanner Power Connection

1. Locate the 24V DC power supply
2. Connect the power supply to the scanner's power input
3. Connect the power supply to mains electricity
4. Verify the scanner powers on (check status LED)

3.2 Network Connection

1. Connect an Ethernet cable from the scanner to the switchbox
2. Connect an Ethernet cable from the robot controller to the switchbox
3. Connect an Ethernet cable from the switchbox to your laptop
4. The devices will auto-configure their network settings

3.3 Robot Connection

The robot controller connects to the measurement application via TCP/IP on **port 5010**. Ensure:

- The laptop and robot are on the same network (or robot connects to laptop's IP)
- Port 5010 is not blocked by firewall
- The robot program is configured with the laptop's IP address

4. Software Installation

4.1 Installing scanCONTROL Configuration Tools

1. Download from the Micro-Epsilon website
2. Run the installer and follow the prompts
3. Restart your computer if prompted

4.2 Installing CloudCompare

1. Visit <https://www.cloudcompare.org/>
2. Download the Windows installer
3. Run the installer and follow the prompts

4.3 Setting Up the Measurement Application

1. Ensure **LLT.dll** is in the parent folder of your executable
2. Place **GetProfiles_Callback_Measure.exe** and **.pdb** in your working folder
3. No installation required - the application runs directly

5. Laser Alignment Procedure

Proper laser alignment is **critical** for accurate measurements. This procedure ensures the laser is correctly positioned over your test area.

5.1 Opening Configuration Tools

1. Ensure the scanner is powered on and connected via Ethernet
2. Launch **scanCONTROL Configuration Tools**
3. The software should auto-detect your scanner
4. Click **Connect** to establish communication

5.2 Selecting the Measurement View

1. In Configuration Tools, navigate to the measurement modes
2. Select **Groove Width/Depth** measurement
3. This view shows a live profile of what the laser sees

5.3 Aligning the Laser

With the live view active, adjust the robot arm position:

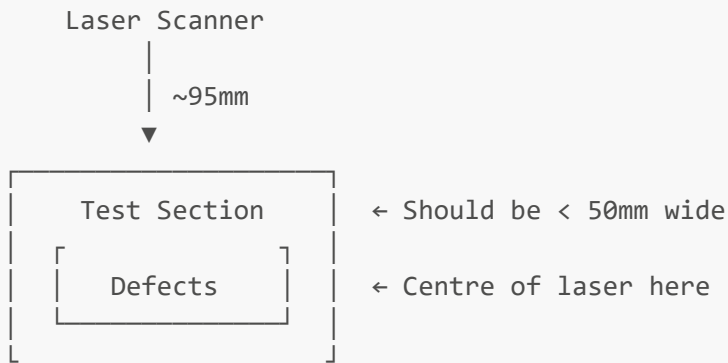
Height Alignment (Z-axis)

- **Target Distance:** Approximately **95mm** from laser to surface
- This is the **median range** of the laser where accuracy is optimal
- The flat section of your test board should be at this distance

- **Tip:** Keep the flat surface distance slightly above 95mm (e.g., 96-97mm). This ensures minor surface deviations remain above the reference plane and are not mistakenly classified as grooves or holes.

Lateral Alignment (X/Y-axis)

- Position the laser so the **centre of the laser line** is over the **centre of your test section**
- The test section should be **under 50mm wide** to fit within the scanner's field of view



5.4 Disconnecting Before Measurement

⚠ Critical Step: You **MUST** disconnect from Configuration Tools before running the measurement application.

1. Click **Disconnect** in Configuration Tools
2. Only then launch the measurement application

Why? The scanner can only communicate with one application at a time. If Configuration Tools is still connected, the measurement application will fail to initialise.

6. Running a Measurement Session

6.1 Launching the Application

1. Navigate to the folder containing `GetProfiles_Callback_Measure.exe`
2. Double-click to run, or open a command prompt and type:

```
GetProfiles_Callback_Measure.exe
```

3. A console window will appear

6.2 Entering Configuration Parameters

The application will prompt you for several parameters. Enter each value and press **Enter**:

```
Enter object length (mm): _  
Enter the distance of the laser scanner from the object (mm): _  
Enter the width needed for each scan section (mm): _  
Enter End-of-Groove Crop (mm): _  
Enter Measurement Window Length (mm, 0 for full): _  
  
--- Analysis Selection ---  
1. Groove Analysis (Standard)  
2. Surface Hole Analysis (Automatic Boundary)  
Selection: _  
  
Enter the depth of the cut (mm): _    ← Only for GROOVE mode
```

See [Section 7](#) for detailed explanations of each parameter.

6.3 Waiting for Robot Connection

After entering parameters, you'll see:

```
>>> WAITING FOR ROBOT CONNECTION <<<  
Waiting for Robot on port 5010...
```

At this point:

1. Start your robot program
2. The robot will connect to the application
3. You'll see: **Robot Connected!**

6.4 During the Scan

The scan runs automatically:

- The robot moves to each segment position
- The scanner captures profile data
- The application analyses each segment
- Progress information is displayed in the console

Do not close the application or disconnect the scanner during scanning.

6.5 Completing the Session

When the scan is complete:

1. You'll see: **Press Enter to exit...**
2. Press **Enter** to close the application
3. Your results are saved in a **Session_YYYY-MM-DD_HH-MM-SS** folder

7. Configuration Parameters Explained

7.1 Object Length (mm)

What it is: The total length of the workpiece to be scanned.

How it's used: The system divides this by the Section Width to determine how many scan segments are needed.

Example: If your test board is 80mm long, enter **80**.

7.2 Laser Distance (mm)

What it is: The distance from the scanner to the surface being measured.

Recommended value: **95mm** (the median range of the scanner)

Why 95mm? This is the optimal working distance where:

- The laser has maximum resolution
- Measurement accuracy is highest
- The field of view is appropriate for typical test sections

How to verify: Use Configuration Tools to check the measured distance to the flat surface of your test piece.

7.3 Scan Section Width (mm)

What it is: How far the robot moves between scan segments.

How it works:

- This value is sent to the robot controller
- The robot moves this distance along the X-axis for each new segment
- Smaller values = more overlap and detail, but longer scan time

Example: For a 80mm object with 15mm section width:

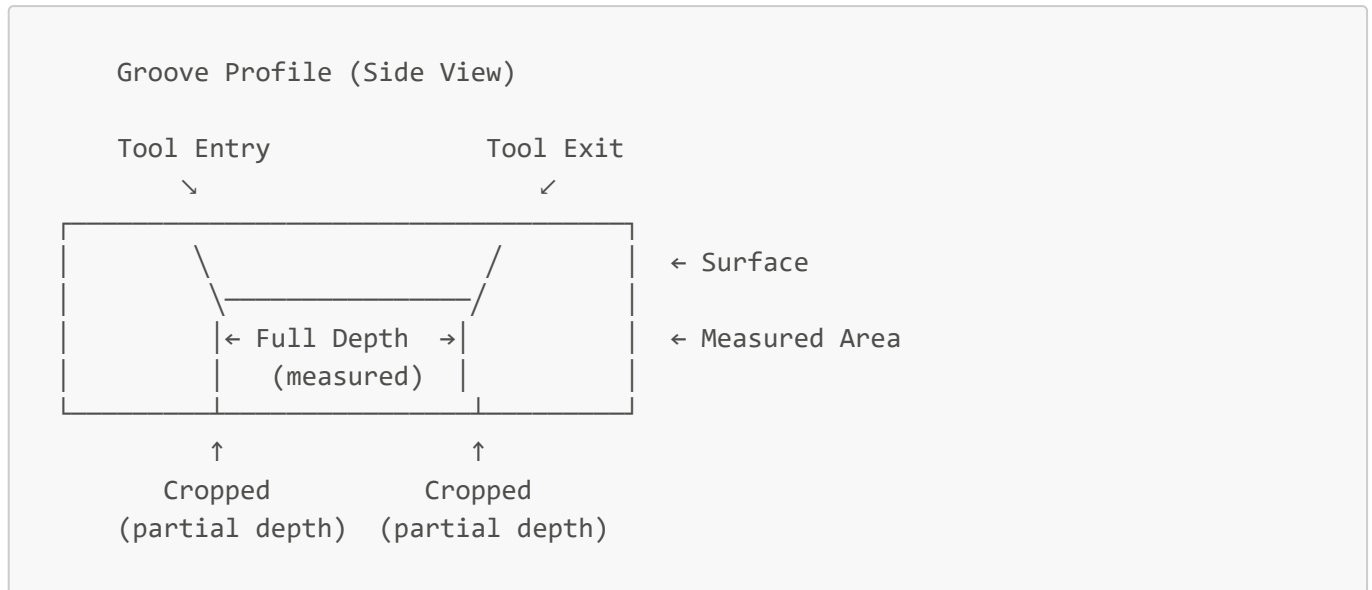
- Number of segments = $80 \div 15 \approx 6$ segments
-

7.4 End-of-Groove Crop (mm)

What it is: Distance to exclude from the ends of detected defects.

Why it's needed: The ends of grooves represent where the cutting tool enters and exits the test material. These entry/exit regions do not represent the full depth of the cut. By cropping these ends, we measure only the portion where the tool was fully submerged in the material.

Typical value: 1-3mm depending on groove characteristics

Visual representation:

7.5 Measurement Window Length (mm)

What it is: Specific length to measure within each defect.

Values:

- 0 = Measure the full length of each detected defect
- >0 = Measure only this specific length (centred on defect)

When to use non-zero: When you need consistent measurement lengths across variable-sized defects.

7.6 Cut Depth (mm) - GROOVE Mode Only

What it is: The expected depth of grooves/cuts in your test piece.

How it's used:

- Sets a "floor" for the analysis
- Points below this depth are excluded
- Prevents erroneous data from affecting measurements

Important: This should be slightly deeper than your actual expected groove depth to ensure all valid points are included.

Example: If your grooves are approximately 1.2mm deep, enter 1.5mm to provide margin.

7.7 Analysis Mode Selection

Mode 1: Groove Analysis (Standard)

Use for: Surface depressions, cuts, scratches, grooves

Detection method: Identifies points that are **below** the reference surface plane

Best for:

- Machined grooves
- Scratch marks
- Surface depressions
- Any defect where material has been removed

Mode 2: Surface Hole Analysis

Use for: Through-holes, slots, gaps in the surface

Detection method: Identifies **missing data** (gaps) in the point cloud

Best for:

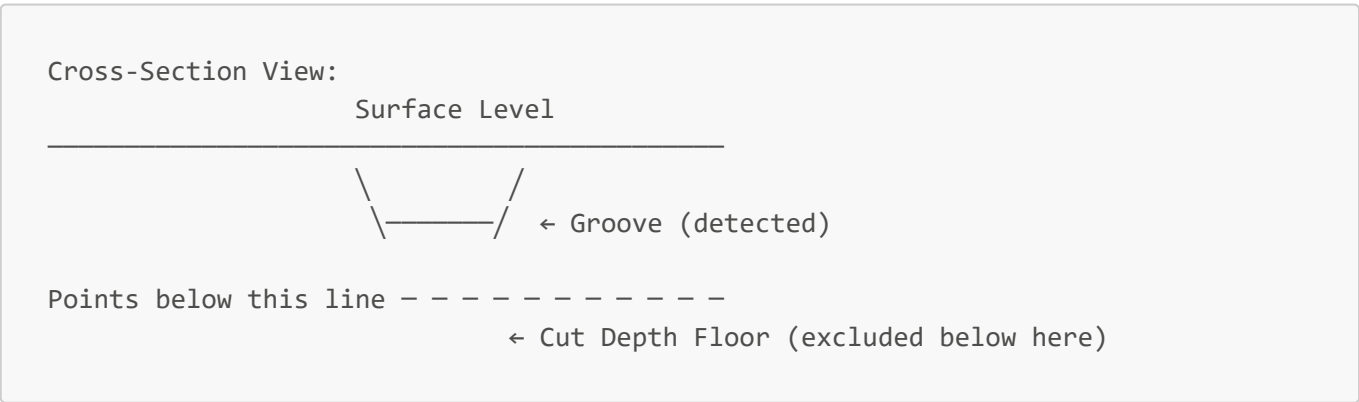
- Drilled holes
- Slots
- Any defect where the laser cannot see a surface (passes through)

8. Understanding Analysis Modes

8.1 Groove Analysis Mode

In Groove mode, the system:

1. **Captures** a 3D point cloud of the surface
2. **Levels** the data to correct for any tilt
3. **Identifies** points below the surface plane
4. **Clusters** connected low points into individual defects
5. **Measures** the area of each defect cluster

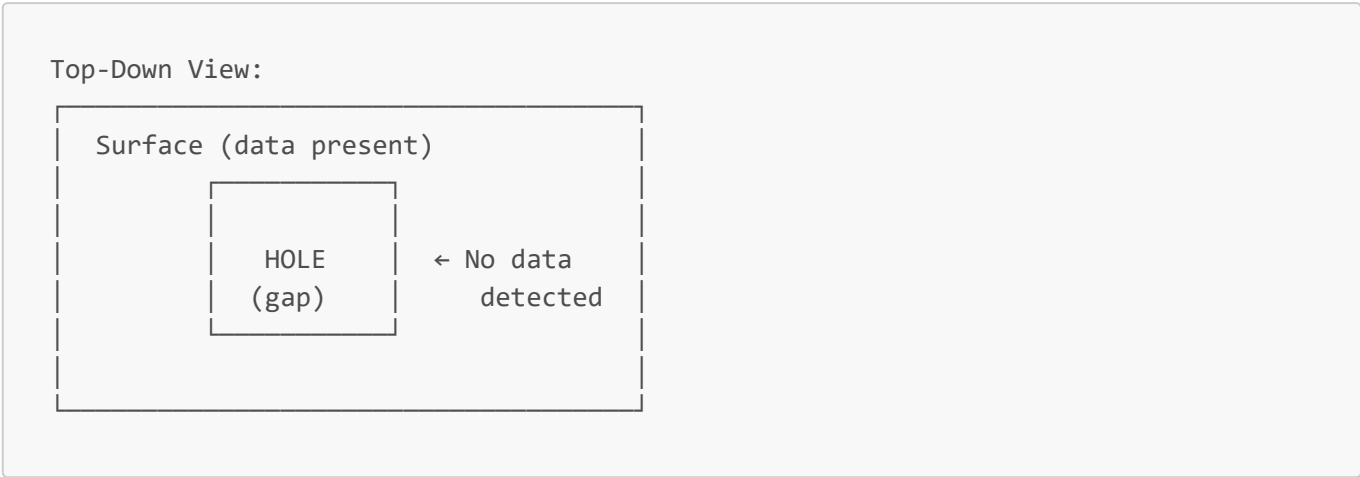


8.2 Hole Analysis Mode

In Hole mode, the system:

1. **Captures** a 3D point cloud of the surface
2. **Identifies** areas where no surface data exists

- 3. **Maps** the boundaries of these gaps
- 4. **Measures** the area of each gap



9. Output Files and Data

9.1 Session Folder

Each measurement session creates a new folder with a timestamp:

```
Session_2026-02-10_14-30-45/
├─ Summary.csv           ← All measurements in tabular format
├─ Segment_0_Raw.ply     ← Raw point cloud (before processing)
├─ Segment_0_Analyzed.ply ← Processed point cloud (colour-coded)
├─ Segment_1_Raw.ply
├─ Segment_1_Analyzed.ply
└─ ...
```

9.2 Summary.csv File

This file contains all measurements in a format suitable for spreadsheet analysis.

Format:

```
Segment_ID,Groove_Index,Area_mm2,Point_Count
0,1,12.3456,5000
0,2,8.7654,3200
1,1,15.0000,6100
```

Columns explained:

Column	Description
Segment_ID	Which scan segment (0, 1, 2, ...)

Column	Description
Groove_Index	Defect number within that segment (1, 2, 3, ...)
Area_mm2	Measured defect area in square millimetres
Point_Count	Number of 3D points in this defect

Opening in Excel:

- 1. Open Excel
- 2. Go to File → Open
- 3. Select the **Summary.csv** file
- 4. The data will import into columns automatically

9.3 PLY Files

PLY (Polygon File Format) files contain 3D point cloud data.

Two types per segment:

File	Contents	Use
Segment_X_Raw.ply	Original captured points (no colour)	Debugging, re-analysis
Segment_X_Analyzed.ply	Processed points with colour coding	Visual inspection

10. Viewing Results in CloudCompare

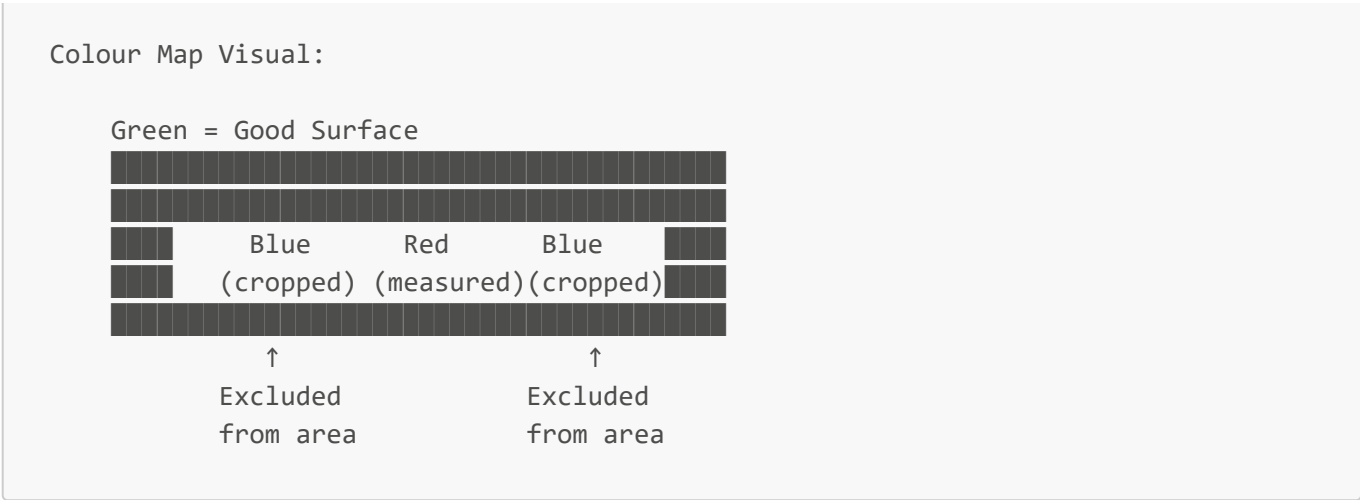
10.1 Opening a PLY File

- 1. Launch **CloudCompare**
- 2. Go to **File** → **Open** (or drag-and-drop the file)
- 3. Select your **_Analyzed.ply** file
- 4. Click **Open**

10.2 Understanding the Colour Coding

The analysed point cloud uses colours to indicate different classifications:

Colour	RGB Value	Meaning
 Green	(0, 255, 0)	Valid surface points - normal surface area
 Red	(255, 0, 0)	Defect points included in measurement
 Blue	(0, 0, 255)	Defect points excluded (cropped region)
 White	(255, 255, 255)	3D text labels showing measurements



10.3 Navigation Controls

Action	Mouse/Keyboard
Rotate view	Left-click + drag
Pan view	Right-click + drag (or Shift + left-click + drag)
Zoom	Scroll wheel
Reset view	Press 1 on keyboard
Top-down view	Press 8 on keyboard

10.4 Adjusting Point Display

If points appear too small or too large:

1. Select your point cloud in the **DB Tree** panel (left side)
2. In the **Properties** panel (typically bottom-left):
 - Find **Point size**
 - Adjust the slider (try 2-5 for most displays)

10.5 Reading 3D Labels

The white 3D text labels show:

- Defect number
- Measured area value

To read labels clearly:

1. Zoom in to the area of interest
2. Rotate to view labels face-on
3. Labels are positioned at the centroid of each defect

11. Interpreting Measurements

11.1 What the Area Measurement Means

The **Area_mm2** value represents the **projected surface area** of the defect, measured in square millimetres.

For grooves, this is the area of the groove as viewed from above (bird's-eye view), not the internal surface area of the groove walls.

11.2 Factors Affecting Measurements

Several factors can influence measurement accuracy:

Factor	Effect	Mitigation
Scanner distance	Affects resolution	Keep at ~95mm
Surface reflectivity	May cause missing points	Clean surface before scanning
Groove depth	Very shallow grooves harder to detect	Ensure Cut Depth is set correctly
Scan speed	Fast motion may reduce point density	Use appropriate robot speed

11.3 Measurement Consistency

For reliable comparisons:

- Use the same parameters for all scans in a test series
- Verify laser alignment before each session
- Check that the 95mm working distance is maintained

11.4 Typical Values

Measurement values vary greatly depending on your specific defects. As a general guide:

Defect Type	Typical Area Range
Fine scratches	0.5 - 2 mm ²
Small grooves	2 - 10 mm ²
Large cuts	10 - 50+ mm ²

Note: These are examples only. Your actual values will depend on your specific test specimens.

12. Troubleshooting

12.1 Connection Issues

"Scanner initialisation failed"

Cause: The scanner couldn't be connected.

Solutions:

1. **Restart the application** - This is common and usually works on the second attempt
2. Ensure Configuration Tools is completely closed
3. Check the Ethernet cable connection
4. Verify the scanner is powered on

"Robot connection timeout"

Cause: The robot didn't connect within the expected time.

Solutions:

1. Check the robot program is running
2. Verify the robot is configured with the correct laptop IP address
3. Check that port 5010 is not blocked by firewall

12.2 No Defects Detected

Possible causes and solutions:

Possible Cause	Solution
Cut Depth too shallow	Increase the Cut Depth value
Wrong analysis mode	Use GROOVE mode for depressions, HOLE mode for through-holes
Surface too reflective	Clean the surface to reduce specular reflections
Scanner not aligned	Re-run the alignment procedure

12.3 Unexpected Area Values

Areas too small or zero

- Check that defects are within the scan area
- Verify the Cut Depth setting isn't too deep
- Ensure the laser distance is approximately 95mm

Areas too large

- May be measuring multiple defects as one
- Check the End-of-Groove Crop setting
- Verify surface levelling is working correctly (check Raw vs Analyzed PLY)

12.4 Recovery from Errors

If an error occurs during scanning:

1. **Close the application** (press Ctrl+C or close the window)
2. **Stop the robot program**
3. **Re-home the robot** to the starting position
4. **Restart the application** from the beginning
5. **Re-enter your parameters** and begin a new session

Note: Partial sessions cannot be resumed. You must restart the entire scan if an error occurs.

12.5 Console Error Messages

Message	Meaning	Action
Initialisation failed	Scanner connection error	Restart application
BUSY responses	Previous segment still processing	Wait - this is normal
ERR response	Analysis error occurred	Check console for details
Timeout	Communication delay	Check network connection

13. Glossary

Term	Definition
Cluster	A group of connected defect points identified as a single defect
Cropping	Excluding edge regions of defects from measurement
Cut Depth	The expected maximum depth of grooves below the surface
Median Range	The optimal scanner working distance (95mm)
PLY	Polygon File Format - a 3D point cloud file format
Point Cloud	A collection of 3D points representing a scanned surface
Profile	A single line of 3D points captured by the laser scanner
RANSAC	Algorithm used to level the surface data
Segment	One section of the total scan area
Session	A complete measurement run from start to finish

Appendix A: Quick Reference Card

Recommended Settings

Parameter	Typical Value
Laser Distance	95mm
Section Width	10-20mm
End-of-Groove Crop	2mm
Measurement Length	0 (full)

Colour Code Summary

Colour	Meaning
 Green	Good surface
 Red	Measured defect
 Blue	Excluded points
 White	Labels

File Locations

What	Where
Application	Your working folder
LLT.dll	Parent folder of application
Results	Session_YYYY-MM-DD_HH-MM-SS/ subfolder

Appendix B: Example Session Walkthrough

Scenario: Measuring 3 Grooves on a Test Board

Test piece: 80mm board with 3 machined grooves (~1.2mm deep)

Step 1: Setup

- Power on scanner (24V)
- Connect Ethernet
- Open Configuration Tools
- Align laser to 95mm, centred on test area
- Disconnect and close Configuration Tools

Step 2: Launch and Configure

Enter object length (mm): 80
Enter the distance of the laser scanner from the object (mm): 95
Enter the width needed for each scan section (mm): 15
Enter End-of-Groove Crop (mm): 2
Enter Measurement Window Length (mm, 0 for full): 0
Selection: 1
Enter the depth of the cut (mm): 1.5

Step 3: Run Scan

- Start robot program
- Wait for "Robot Connected!"
- Scan runs automatically (~2-3 minutes)
- "Press Enter to exit..."

Step 4: Review Results

Open: Session_2026-02-10_14-30-45/

Summary.csv shows:

Segment_ID,Groove_Index,Area_mm2,Point_Count

0,1,8.234,3200

2,1,7.891,3100

4,1,8.102,3150

View Segment_2_Analyzed.ply in CloudCompare:

- Green surface with red groove visible
- White label showing "1: 7.89 mm²"

End of User Manual

For technical support or questions about this system, contact the development team.